Name: _____ Date: _____

UNIT 6 · FUNCTIONS · GUIDED NOTES

Tables & Systems of Equations

Fill in the blanks. Practice problems at the end.

1 Writing Linear Equations from a Table

Every linear equation can be written as y = ____ x + ____ , where m is the _____ (rate of change) and b is the _____ (the y-value when x = 0).

$$m = (y_2 - y_1) \div (x_2 - x_1) = \text{change in } y \div \text{change in } x$$

Example 1 · Table includes x = 0

x	y
0	3
1	7
2	11
3	15

Step 1. From (0, 3) to (1, 7):

change in y = ___, change in x = ___

slope m = ___

Step 2. When x = 0, y = ___. So b = ___.

Step 3. Equation: y = _____ x + _____

Example 2 · No x = 0 row — work backwards for b

x	y
2	5
4	11
6	17
8	23

Step 1. From (2, 5) to (4, 11):

change in y = ___, change in x = ___. m = ___.

Step 2. Use b = y − m·x. Pick (2, 5):

b = ___ − (___)(___) = ___

Step 3. Equation: y = _____ x + _____

> **Shortcut for finding b**
> Once you know m, pick any row (x, y). Then: b = y − m · x

2 Solving Systems of Equations by Graphing

A system of equations is ___ or more equations that use the same _____. In 8th grade, we work with systems of two linear equations in x and y.

Graphing Method

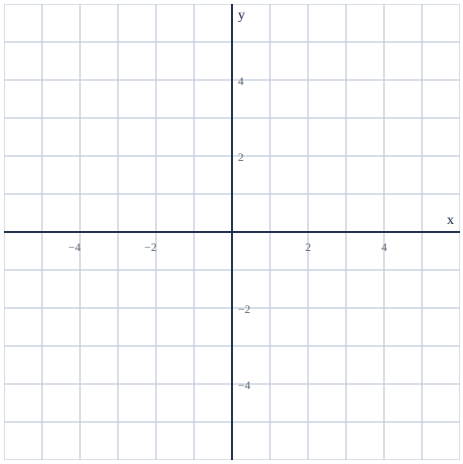
Graph each line on the same coordinate plane. The solution is the _____ point.

Steps

1. Graph the first line (using slope and y-intercept).
2. Graph the second line on the same plane.
3. Find where they _____. That ordered pair is the solution.
4. Check by plugging the point into both equations.

Example · Graph and solve

EQ 1	y = x + 2
EQ 2	y = −x + 4



EQ 1: y = x + 2

slope = ___, y-int = ___

EQ 2: y = −x + 4

slope = ___, y-int = ___

Solution (intersection):

(_____ , _____)

Check:

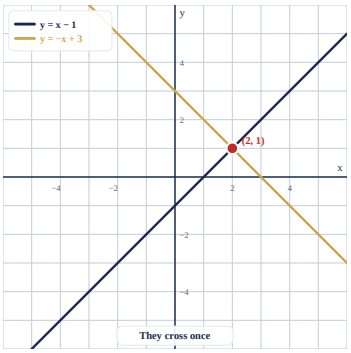
EQ 1: ___ = ___ + 2

EQ 2: ___ = −___ + 4

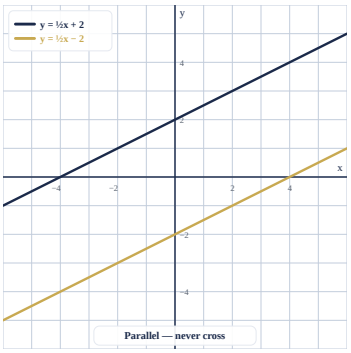
Three Possible Outcomes

Case	What You See	How Many Solutions?
1	Lines cross at one point	_____ solution
2	Lines are _____ (never cross)	_____ solutions
3	Both equations graph the same line	_____ solutions

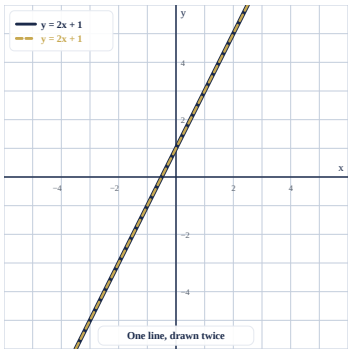
CASE 1



CASE 2



CASE 3



Count the intersection points: one point → one solution, zero points → no solution, every point → infinitely many.

3 What Does "Solution" Actually Mean?

A solution to a system of equations is an ordered pair (x, y) that makes _____ equations _____ at the same time.

Two ways to say it

- **Graphically:** the solution is the _____ where the lines cross.
- **Algebraically:** the solution is the (x, y) that, when plugged in, makes _____ equations true.

Testing Whether a Point Is a Solution

Plug x and y into _____ equation. If the point makes _____ equations true, it is a solution. If it fails even one, it is not a solution.

Example · Is (2, 5) a solution?

EQ 1	y = 2x + 1
EQ 2	y = 3x − 1

1. Test EQ 1 with x = 2, y = 5: ___ = 2(___) + 1 = ___. True? _____
2. Test EQ 2 with x = 2, y = 5: ___ = 3(___) − 1 = ___. True? _____
3. Both true, so (2, 5) _____ a solution.

Takeaway

You cannot tell just by looking at a point. Always test _____ equations.

4 Solving Systems by Substitution

Graphing shows the solution but is slow and hard to read when answers aren't whole numbers. Substitution is algebraic — it always gives an exact answer.

Substitution procedure

1. One equation should be solved for a variable (usually y = ...).
2. Substitute the expression into the _____ equation.
3. Solve that equation for ___.
4. Plug that value back in to find the _____ variable.
5. Write the solution as an _____ pair (x, y) and check.

Example 1 · Both equations in y = form

EQ 1	y = 2x + 3
EQ 2	y = −x + 6

1. Both equations equal y, so they equal each other: _____ = −x + 6
2. Solve for x. Add x: 3x + 3 = 6. Subtract 3: 3x = ___. Divide: x = ___.
3. Back-substitute into EQ 1: y = 2(___) + 3 = ___.
4. Solution: (___ , ___)

Example 2 · One equation in y = form, one not

EQ 1	y = 3x − 1
EQ 2	2x + y = 9

1 EQ 1 is already solved for y: y = _____

2 Substitute (_____) for y in EQ 2: 2x + (_____) = 9

3 Combine like terms: ___x − 1 = 9. Add 1: 5x = ___. Divide: x = ___.

4 Back-substitute: y = 3(___) − 1 = ___.

5 Solution: (___ , ___)

Practice Problems

Show your work. Check your answers.

1. Write the linear equation for this table.

x	y
0	−4
1	−1
2	2
3	5

slope m = ___

y-intercept b = ___

Equation: y = _____ x + _____

2. Write the linear equation for this table. (no x = 0 row)

x	y
2	7
4	13
6	19
8	25

slope m = ___

Use b = y − m·x with (___, ___): b = ___ − (___)(___) = ___

Equation: y = _____ x + _____

3. Is (3, 5) a solution to this system? Show your work. Circle YES or NO.

EQ 1	y = 2x − 1
EQ 2	y = −x + 8

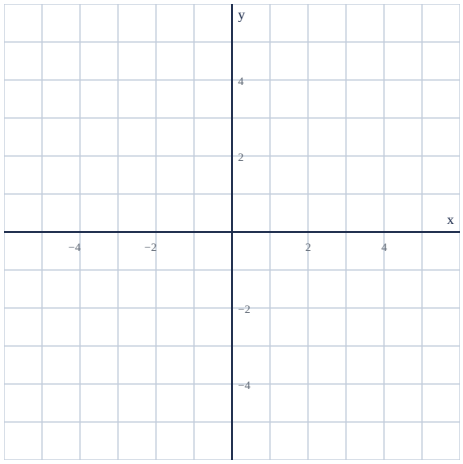
EQ 1 test: ___ = 2(___) − 1 = ___ True or False? _____

EQ 2 test: ___ = −(___) + 8 = ___ True or False? _____

Final answer (circle): YES / NO

4. Solve this system by graphing on the grid below.

EQ 1	y = x + 1
EQ 2	y = −x + 5



EQ 1 slope = ___, y-int = ___

EQ 2 slope = ___, y-int = ___

Solution: (____ , ____)

Check:

EQ 1: ___ = ___ + 1

EQ 2: ___ = −___ + 5

5. Solve by substitution.

EQ 1	y = x + 3
EQ 2	y = 2x − 1

Set equal: _____ = 2x − 1

Solve for x: x = ___

Find y: y = ___ + 3 = ___

Solution: (____ , ____)

6. Solve by substitution. *(One equation needs a quick substitution.)*

EQ 1	y = 3x + 2
EQ 2	x + y = 10

Substitute (_____) for y in the second equation: x + (_____) = 10

Combine: ___x + ___ = 10

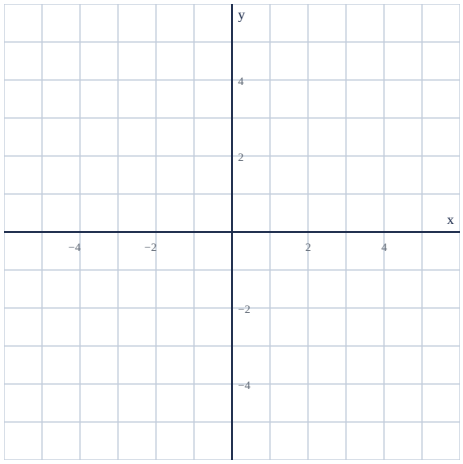
Solve for x: x = ___

Find y: y = 3(___) + 2 = ___

Solution: (____ , ____)

7. Solve this system by graphing on the grid below.

EQ 1	$y = 2x - 3$
EQ 2	$y = x + 1$



EQ 1 slope = ___, y-int = ___

EQ 2 slope = ___, y-int = ___

Solution: (____ , ____)

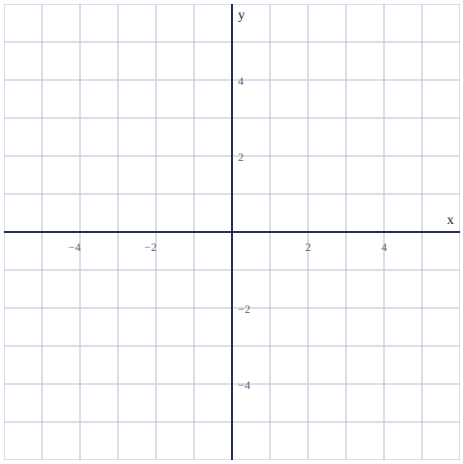
Check:

EQ 1: ___ = 2(___) − 3

EQ 2: ___ = ___ + 1

8. Solve this system by graphing on the grid below.

EQ 1	$y = \frac{1}{3}x + 1$
EQ 2	$y = -x + 5$



EQ 1 slope = ___, y-int = ___

EQ 2 slope = ___, y-int = ___

Solution: (____ , ____)

Check:

EQ 1: ___ = $\frac{1}{3}$(___) + 1

EQ 2: ___ = −___ + 5

9. One, none, or infinitely many? Compare the slopes and y-intercepts first — you do not need to graph these.

a) $y = 2x + 1$ and $y = 2x - 4$ → slopes ____ and ____ , y-intercepts ____ and ____ → _____ solution(s)

b) $y = 3x - 2$ and $y = -x + 2$ → slopes ____ and ____ , y-intercepts ____ and ____ → _____ solution(s)

c) $y = -\frac{1}{2}x + 3$ and $y = -\frac{1}{2}x + 3$ → slopes ____ and ____ , y-intercepts ____ and ____ → _____ solution(s)

> **Remember**
> Different slopes → the lines cross once. Same slope, different y-intercept → parallel, never cross. Same slope *and* same y-intercept → one line written twice.